Bayesian Optimization Report: Function 6

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.675	Medium
2	0.673	Medium
3	0.674	Medium
4	0.674	Medium
5	0.674	Medium
6	0.673	Medium
7	0.673	Medium
8	0.674	Medium
9	0.676	Medium
10	0.677	Medium
11	0.678	Medium
12	0.679	Medium
13	0.681	Medium
14	0.683	Medium
15	0.685	Medium
16	0.686	Medium
17	0.688	Medium
18	0.687	Medium
19	0.689	Medium
20	0.693	Medium
21	0.695	Medium
22	0.695	Medium
23	0.698	Medium
24	0.699	Medium
25	0.701	Good
26	0.704	Good
27	0.705	Good
28	0.707	Good
29	0.709	Good
30	0.710	Good

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.1479	0.0340	0.3254
2	0.1434	0.0472	0.2765
3	0.1183	0.0386	0.2820
4	0.1319	0.0310	0.2574
5	0.1047	0.0239	0.2369
6	0.0955	0.0247	0.2421
7	0.1938	0.0827	0.3891
8	0.1999	0.0860	0.4319
9	0.1975	0.0800	0.3759
10	0.1889	0.0713	0.4112
11	0.1848	0.0660	0.3916
12	0.1902	0.0792	0.3938
13	0.0595	0.0157	0.1989
14	0.0609	0.0147	0.2513
15	0.0596	0.0164	0.1588
16	0.0615	0.0160	0.2052
17	0.0612	0.0172	0.2013
18	0.0642	0.0165	0.2499
19	0.0598	0.0156	0.2364
20	0.0578	0.0158	0.2140
21	0.0535	0.0165	0.2045
22	0.0580	0.0149	0.2366
23	0.0552	0.0149	0.1597
24	0.0561	0.0156	0.1867
25	0.0571	0.0145	0.2840
26	0.0559	0.0146	0.2125
27	0.0538	0.0146	0.1795
28	0.0511	0.0146	0.1408
29	0.0518	0.0145	0.1750
30	0.0548	0.0155	0.1622

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.44336687 0.37634478 0.61577895 0.81536759 0.20228001]	-0.14577131528243403	Matern(ν=2.5)	PI	0.675	None
2	[0.50315847 0.39779885 0.58977416 0.87172732 0.12872793]	-0.1364127329125484	Matern(ν=1.5)	PI	0.673	None
3	[0.49449958 0.35294965 0.61347754 0.86809096 0.15714604]	-0.11026111952729223	Matern(ν=2.5)	PI	0.674	None
4	[0.49884673 0.2120813 0.5838569 0.87937179 0.15523505]	-0.08569658134581037	Matern(ν=1.5)	PI	0.674	None
5	[0.51813184 0.28043269 0.53901962 0.93035789 0.20180744]	-0.058257417123891875	Matern(ν=2.5)	PI	0.674	None
6	[0.5234979 0.20823502 0.59960599 0.98398163 0.20072037]	-0.0635885011745212	Matern(ν=2.5)	PI	0.673	None
7	[0.48391121 0.21730278 0.5828994 1. 0.22821987]	-0.05160741343583508	Matern(ν=0.5)	PI	0.673	None
8	[0.51042552 0.20910161 0.55320474 0.97527005 0.20930649]	-0.05422170338294818	Matern(ν=0.5)	PI	0.674	None
9	[0.47210045 0.2651937 0.58547853 1. 0.19654071]	-0.057243614713832436	Matern(ν=0.5)	PI	0.676	None
10	[0.48480472 0.30588212 0.54874586 1. 0.22945162]	-0.056849257976577805	Matern(ν=0.5)	PI	0.677	None
11	[0.48767774 0.28433582 0.55188717 1. 0.20795292]	-0.052426878518076814	Matern(ν=0.5)	PI	0.678	None
12	[0.46124803 0.21315153 0.55640452 1. 0.19223094]	-0.06204512686352248	Matern(ν=0.5)	PI	0.679	None
13	[0.48720826 0.27486899 0.56262445 1. 0.23275786]	-0.05326806774791537	Matern(ν=2.5)	PI	0.681	None
14	[0.48676266 0.27806687 0.57188436 0.99540283 0.21638775]	-0.0513046305443281	Matern(ν=2.5)	PI	0.683	None
15	[0.49890776 0.32279367 0.539092 1. 0.19694975]	-0.05872246197945996	Matern(ν=2.5)	PI	0.685	None
16	[0.51677292 0.26855909 0.54353597 0.97709503 0.20225703]	-0.05526356014239142	Matern(ν=2.5)	PI	0.686	None
17	[0.49313421 0.18469711 0.54775602 0.99093634 0.23120867]	-0.057966163343194044	Matern(ν=2.5)	PI	0.688	None

18	[0.51362446 0.18582087 0.57591888 1. 0.22513017]	-0.05610338788 6804246	Matern(ν=1.5)	PI	0.687	None
19	[0.51710249 0.25524209 0.58227946 1. 0.22808633]	-0.05573856566 287749	Matern(ν=1.5)	PI	0.689	None
20	[0.50507964 0.20154589 0.60252837 0.94396719 0.24307043]	-0.06309543752 007962	Matern(ν=2.5)	PI	0.693	None
21	[0.50832491 0.27906191 0.58763769 1. 0.24590948]	-0.06133264949 334194	Matern(ν=2.5)	PI	0.695	None
22	[0.49437986 0.22984873 0.54137874 0.95929516 0.20621043]	-0.05386066522 601096	Matern(ν=1.5)	PI	0.695	None
23	[0.50063989 0.31489145 0.55706823 0.97820369 0.20196997]	-0.05482838695 108927	Matern(ν=2.5)	PI	0.698	None
24	[0.47613145 0.28610721 0.55809357 1. 0.18687822]	-0.05930933744 466249	Matern(ν=1.5)	PI	0.699	None
25	[0.50564536 0.24650206 0.55709556 1. 0.20952806]	-0.05140076929 904347	Matern(ν=1.5)	PI	0.701	None
26	[0.49577538 0.23110381 0.56409672 0.97388851 0.24305191]	-0.05529595209 4069456	Matern(ν=2.5)	PI	0.704	None
27	[0.51987248 0.24828665 0.58023281 1. 0.21568458]	-0.05488810819 272061	Matern(ν=2.5)	PI	0.705	None
28	[0.48712224 0.22630124 0.55289734 1. 0.19979773]	-0.05282340288 68817	Matern(ν=2.5)	PI	0.707	None
29	[0.48216407 0.2805553 0.57784087 1. 0.22385156]	-0.05382334868 697303	Matern(ν=2.5)	PI	0.709	None
30	[0.51553236 0.21261591 0.54045811 0.9592989 0.22077967]	-0.05738788279 606788	Matern(ν=2.5)	PI	0.710	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	1.845587
EI	0.047119
PI	0.486275
THOMPSON	1.628620
KG	0.047119
ENTROPY	3.389959
PORTFOLIO	1.552166

Iteration 2

Method	Max Value
UCB	1.546955
EI	0.036174
PI	0.373662
THOMPSON	1.570778
KG	0.036174
ENTROPY	2.926736
PORTFOLIO	1.314052

Iteration 3

Method	Max Value
UCB	1.639187
EI	0.077196
PI	0.587036
THOMPSON	1.627937
KG	0.077196
ENTROPY	3.388668
PORTFOLIO	1.489023

Iteration 4

Method	Max Value
UCB	1.634830
EI	0.054626
PI	0.480517
THOMPSON	1.639236
KG	0.054626
ENTROPY	3.305721
PORTFOLIO	1.421230

Iteration 5

Method	Max Value
UCB	1.613528
EI	0.036499

PI	0.533446
THOMPSON	1.680519
KG	0.036499
ENTROPY	3.097185
PORTFOLIO	1.261416

Iteration 6

Method	Max Value
UCB	1.597735
EI	0.010160
PI	0.223533
THOMPSON	1.617445
KG	0.010160
ENTROPY	2.897539
PORTFOLIO	1.248588

Iteration 7

Method	Max Value
UCB	1.667738
EI	0.033709
PI	0.345631
THOMPSON	1.789247
KG	0.033709
ENTROPY	3.848185
PORTFOLIO	1.491008

Iteration 8

Method	Max Value
UCB	1.656164
EI	0.023618
PI	0.289143
THOMPSON	1.700363
KG	0.023618
ENTROPY	3.774416
PORTFOLIO	1.437801

Iteration 9

Method	Max Value
UCB	1.651797
EI	0.022670
PI	0.281201
THOMPSON	1.853981
KG	0.022670
ENTROPY	3.615864
PORTFOLIO	1.397551

Iteration 10

Method	Max Value
UCB	1.650750
EI	0.021853
PI	0.284873
THOMPSON	1.869018
KG	0.021853
ENTROPY	3.821604
PORTFOLIO	1.313577

Iteration 11

Method	Max Value
UCB	1.655201
EI	0.020456
PI	0.291265
THOMPSON	1.789273
KG	0.020456
ENTROPY	3.751302
PORTFOLIO	1.324917

Iteration 12

Method	Max Value
UCB	1.651275
EI	0.018629

PI	0.244938
THOMPSON	1.790548
KG	0.018629
ENTROPY	3.911385
PORTFOLIO	1.306754

Iteration 13

Method	Max Value
UCB	1.591935
EI	0.001157
PI	0.132899
THOMPSON	1.589504
KG	0.001157
ENTROPY	2.551199
PORTFOLIO	1.069205

Iteration 14

Method	Max Value
UCB	1.598403
EI	0.001584
PI	0.178491
THOMPSON	1.593430
KG	0.001584
ENTROPY	2.658946
PORTFOLIO	1.106173

Iteration 15

Method	Max Value
UCB	1.595692
EI	0.000518
PI	0.059238
THOMPSON	1.592977
KG	0.000518
ENTROPY	2.611711
PORTFOLIO	1.084027

Iteration 16

Method	Max Value
UCB	1.602416
EI	0.000692
PI	0.086816
THOMPSON	1.598499
KG	0.000692
ENTROPY	2.570201
PORTFOLIO	1.076052

Iteration 17

Method	Max Value
UCB	1.605881
EI	0.000742
PI	0.077409
THOMPSON	1.607404
KG	0.000742
ENTROPY	2.604128
PORTFOLIO	1.082813

Iteration 18

Method	Max Value
UCB	1.603755
EI	0.000686
PI	0.081969
THOMPSON	1.593226
KG	0.000686
ENTROPY	2.597787
PORTFOLIO	1.073867

Iteration 19

Method	Max Value
UCB	1.608759
EI	0.000708

PI	0.085465
THOMPSON	1.615885
KG	0.000708
ENTROPY	2.611957
PORTFOLIO	1.083152

Iteration 20

Method	Max Value
UCB	1.612280
EI	0.000377
PI	0.039976
THOMPSON	1.622451
KG	0.000377
ENTROPY	2.615612
PORTFOLIO	1.080879

Iteration 21

Method	Max Value
UCB	1.613608
EI	0.000135
PI	0.021055
THOMPSON	1.610627
KG	0.000135
ENTROPY	2.580167
PORTFOLIO	1.083295

Iteration 22

Method	Max Value
UCB	1.623968
EI	0.000898
PI	0.110763
THOMPSON	1.617917
KG	0.000898
ENTROPY	2.588601
PORTFOLIO	1.089403

Iteration 23

Method	Max Value
UCB	1.629865
EI	0.000625
PI	0.084941
THOMPSON	1.629616
KG	0.000625
ENTROPY	2.597690
PORTFOLIO	1.093418

Iteration 24

Method	Max Value
UCB	1.625495
EI	0.000194
PI	0.030307
THOMPSON	1.641859
KG	0.000194
ENTROPY	2.640573
PORTFOLIO	1.086620

Iteration 25

Method	Max Value
UCB	1.640612
EI	0.001438
PI	0.168537
THOMPSON	1.637231
KG	0.001438
ENTROPY	2.624624
PORTFOLIO	1.093548

Iteration 26

Method	Max Value
UCB	1.641977
EI	0.000622

PI	0.082278
THOMPSON	1.640669
KG	0.000622
ENTROPY	2.628294
PORTFOLIO	1.096280

Iteration 27

Method	Max Value
UCB	1.646349
EI	0.000662
PI	0.087789
THOMPSON	1.640917
KG	0.000662
ENTROPY	2.644101
PORTFOLIO	1.104056

Iteration 28

Method	Max Value
UCB	1.652811
EI	0.001022
PI	0.128960
THOMPSON	1.652223
KG	0.001022
ENTROPY	2.645162
PORTFOLIO	1.103807

Iteration 29

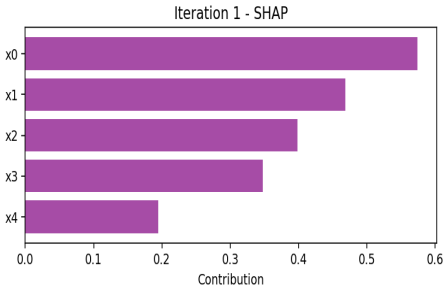
Method	Max Value
UCB	1.655259
EI	0.000769
PI	0.103202
THOMPSON	1.648085
KG	0.000769
ENTROPY	2.637614
PORTFOLIO	1.100240

Iteration 30

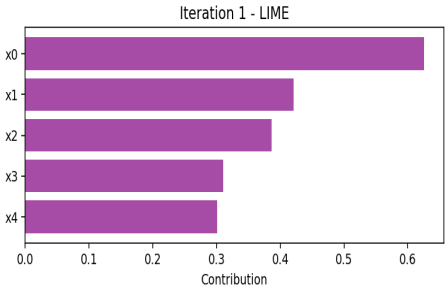
Method	Max Value
UCB	1.654755
EI	0.000345
PI	0.050283
THOMPSON	1.650334
KG	0.000345
ENTROPY	2.647090
PORTFOLIO	1.103738

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



Iteration 1 - SHAP Values

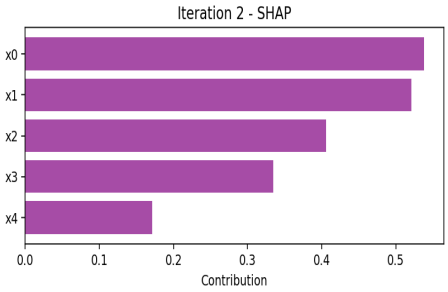
Feature	SHAP Value
x0	0.57396
x1	0.46874
x2	0.39842
x3	0.34800

x4	0.19461

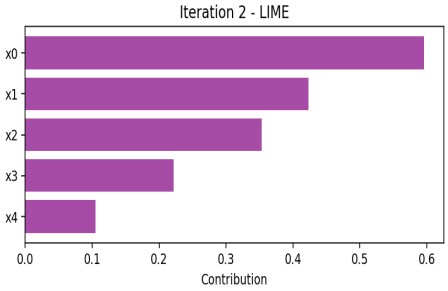
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.62535
x1	0.42071
x2	0.38640
x3	0.31018
x4	0.30111

Iteration 2 - SHAP



Iteration 2 - LIME



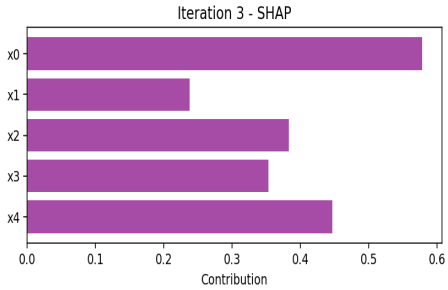
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	0.53795
x1	0.52133
x2	0.40597
x3	0.33467
x4	0.17095

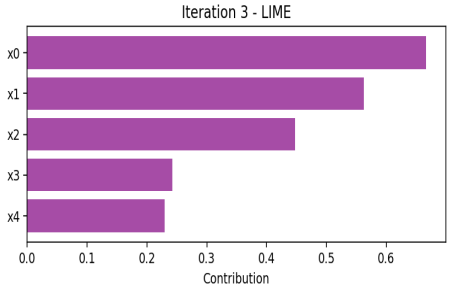
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.59557
x1	0.42285
x2	0.35361
x3	0.22197
x4	0.10521

Iteration 3 - SHAP



Iteration 3 - LIME



Iteration 3 - SHAP Values

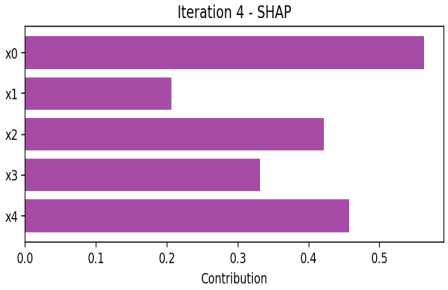
Feature	SHAP Value
x0	0.57800
x1	0.23723
x2	0.38285
x3	0.35306
x4	0.44665

Iteration 3 - LIME Values

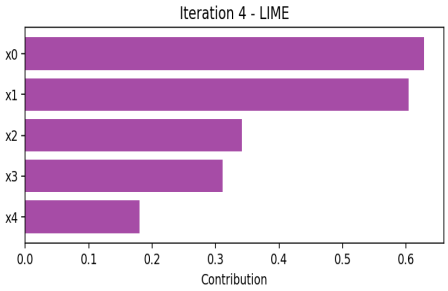
Feature	LIME Value
x0	0.66641
x1	0.56280

x2	0.44805
x3	0.24293
x4	0.22960

Iteration 4 - SHAP



Iteration 4 - LIME



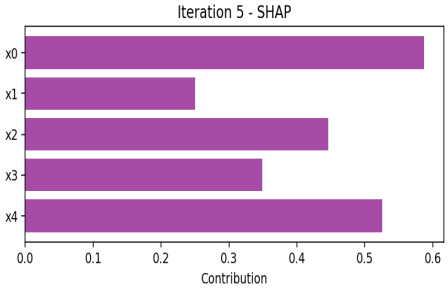
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	0.56238
x1	0.20613
x2	0.42144
x3	0.33085
x4	0.45672

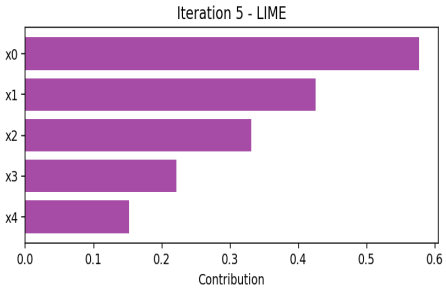
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.62830
x1	0.60407
x2	0.34122
x3	0.31116
x4	0.17981

Iteration 5 - SHAP



Iteration 5 - LIME



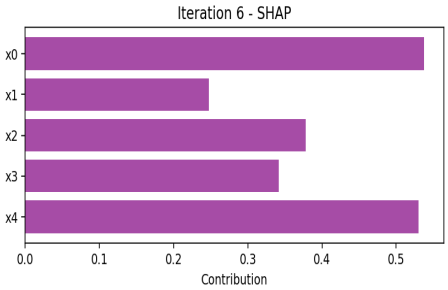
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	0.58694
x1	0.25026
x2	0.44597
x3	0.34895
x4	0.52546

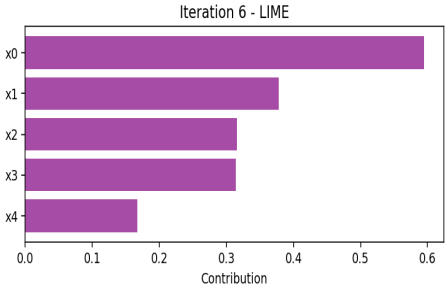
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.57662
x1	0.42514
x2	0.33063
x3	0.22157
x4	0.15238

Iteration 6 - SHAP



Iteration 6 - LIME



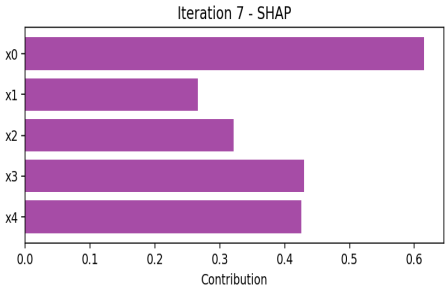
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	0.53733
x1	0.24720
x2	0.37840
x3	0.34189
x4	0.53017

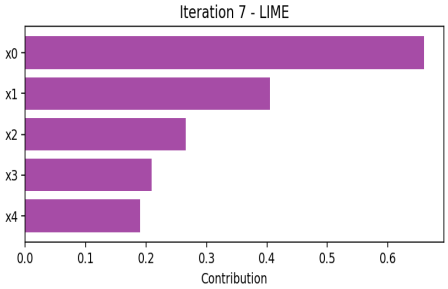
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.59447
x1	0.37805
x2	0.31550
x3	0.31377
x4	0.16688

Iteration 7 - SHAP



Iteration 7 - LIME



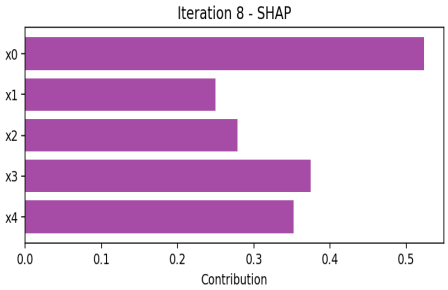
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	0.61462
x1	0.26629
x2	0.32159
x3	0.42978
x4	0.42586

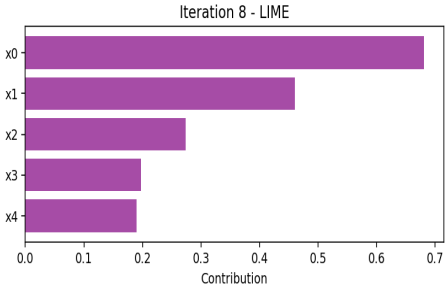
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.65979
x1	0.40462
x2	0.26598
x3	0.20906
x4	0.18987

Iteration 8 - SHAP



Iteration 8 - LIME



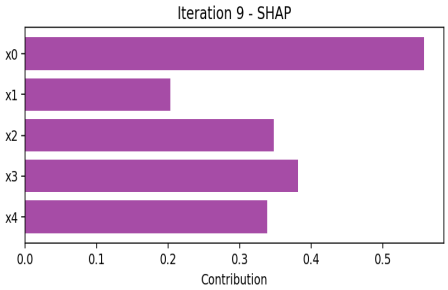
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.52311
x1	0.24987
x2	0.27868
x3	0.37458
x4	0.35231

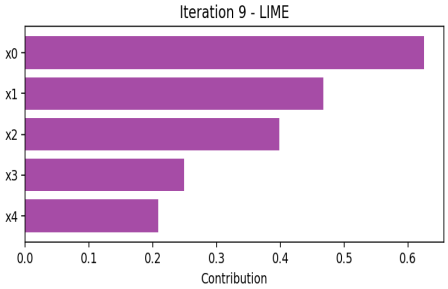
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.68044
x1	0.46000
x2	0.27348
x3	0.19716
x4	0.18969

Iteration 9 - SHAP



Iteration 9 - LIME



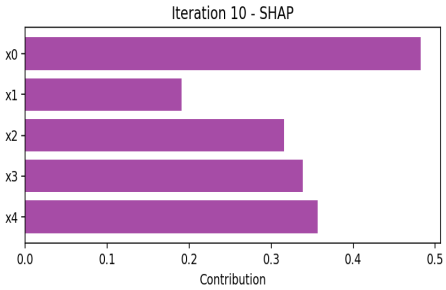
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	0.55734
x1	0.20245
x2	0.34758
x3	0.38105
x4	0.33863

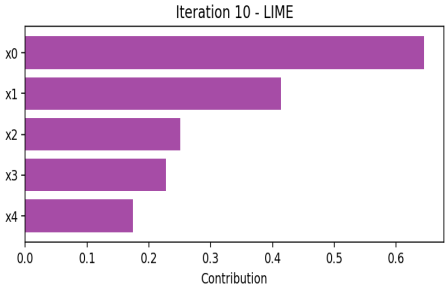
Iteration 9 - LIME Values

Feature	LIME Value
x0	0.62507
x1	0.46701
x2	0.39864
x3	0.24938
x4	0.20832

Iteration 10 - SHAP



Iteration 10 - LIME



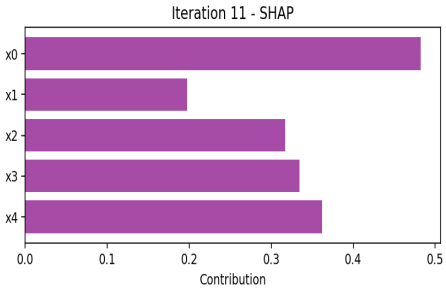
Iteration 10 - SHAP Values

Feature	SHAP Value
x0	0.48221
x1	0.19041
x2	0.31587
x3	0.33838
x4	0.35659

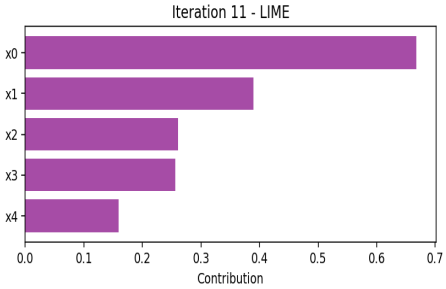
Iteration 10 - LIME Values

Feature	LIME Value
x0	0.64486
x1	0.41384
x2	0.25035
x3	0.22728
x4	0.17410

Iteration 11 - SHAP



Iteration 11 - LIME



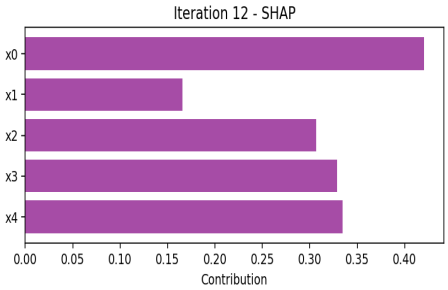
Iteration 11 - SHAP Values

Feature	SHAP Value
x0	0.48211
x1	0.19703
x2	0.31683
x3	0.33457
x4	0.36220

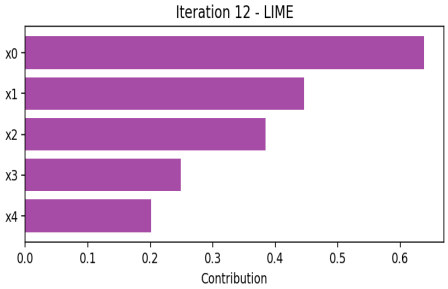
Iteration 11 - LIME Values

Feature	LIME Value
x0	0.66809
x1	0.38923
x2	0.26056
x3	0.25606
x4	0.15959

Iteration 12 - SHAP



Iteration 12 - LIME



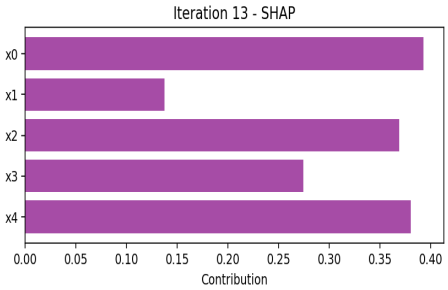
Iteration 12 - SHAP Values

Feature	SHAP Value
x0	0.42033
x1	0.16569
x2	0.30703
x3	0.32897
x4	0.33440

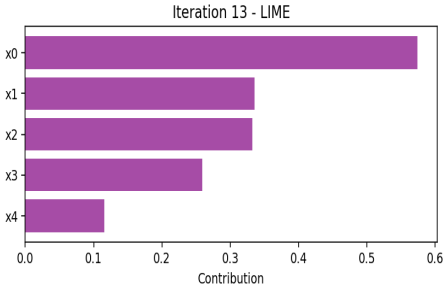
Iteration 12 - LIME Values

Feature	LIME Value
x0	0.63800
x1	0.44620
x2	0.38467
x3	0.24857
x4	0.20146

Iteration 13 - SHAP



Iteration 13 - LIME



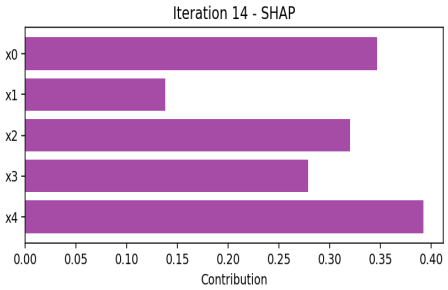
Iteration 13 - SHAP Values

Feature	SHAP Value
x0	0.39312
x1	0.13756
x2	0.36893
x3	0.27432
x4	0.38035

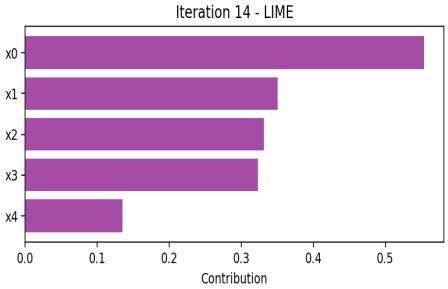
Iteration 13 - LIME Values

Feature	LIME Value
x0	0.57425
x1	0.33558
x2	0.33253
x3	0.25939
x4	0.11593

Iteration 14 - SHAP



Iteration 14 - LIME



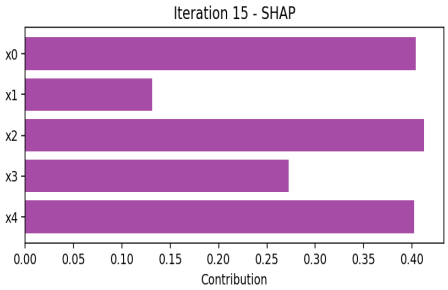
Iteration 14 - SHAP Values

Feature	SHAP Value
x0	0.34643
x1	0.13801
x2	0.31985
x3	0.27840
x4	0.39192

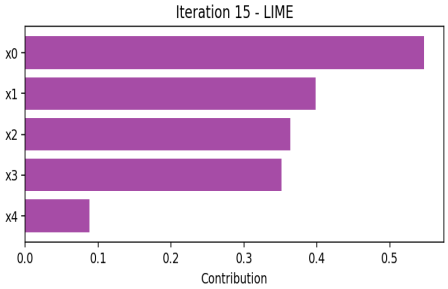
Iteration 14 - LIME Values

Feature	LIME Value
x0	0.55359
x1	0.35050
x2	0.33169
x3	0.32322
x4	0.13507

Iteration 15 - SHAP



Iteration 15 - LIME



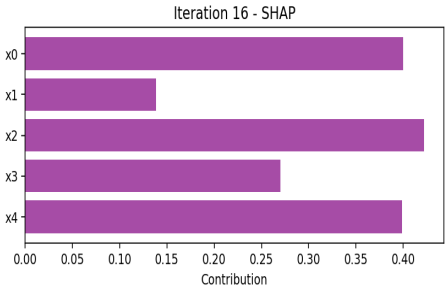
Iteration 15 - SHAP Values

Feature	SHAP Value
x0	0.40414
x1	0.13111
x2	0.41230
x3	0.27235
x4	0.40195

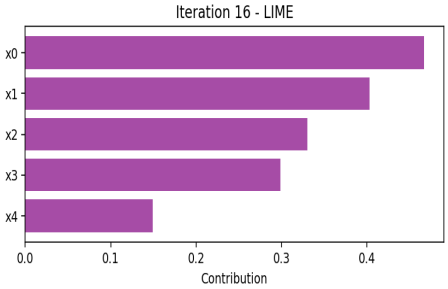
Iteration 15 - LIME Values

Feature	LIME Value
x0	0.54693
x1	0.39807
x2	0.36369
x3	0.35125
x4	0.08812

Iteration 16 - SHAP



Iteration 16 - LIME



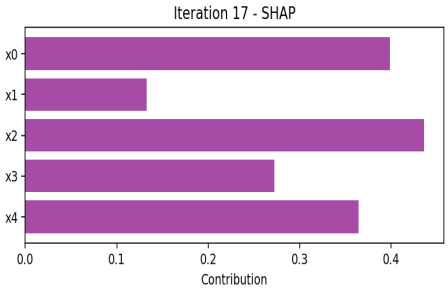
Iteration 16 - SHAP Values

Feature	SHAP Value
x0	0.40016
x1	0.13853
x2	0.42195
x3	0.26985
x4	0.39876

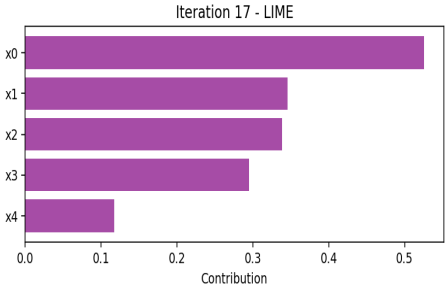
Iteration 16 - LIME Values

Feature	LIME Value
x0	0.46683
x1	0.40309
x2	0.33065
x3	0.29852
x4	0.14927

Iteration 17 - SHAP



Iteration 17 - LIME



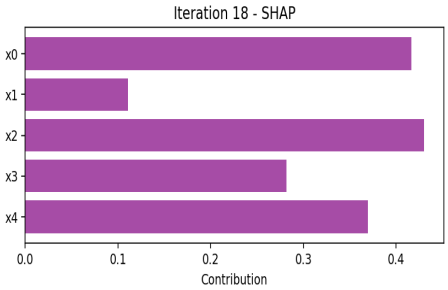
Iteration 17 - SHAP Values

Feature	SHAP Value
x0	0.39846
x1	0.13260
x2	0.43587
x3	0.27244
x4	0.36437

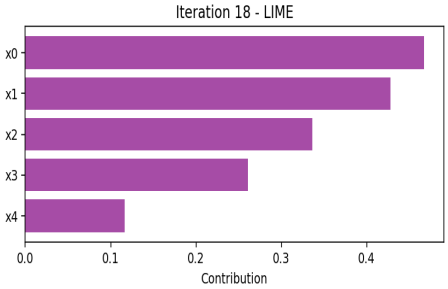
Iteration 17 - LIME Values

Feature	LIME Value
x0	0.52553
x1	0.34600
x2	0.33844
x3	0.29465
x4	0.11685

Iteration 18 - SHAP



Iteration 18 - LIME



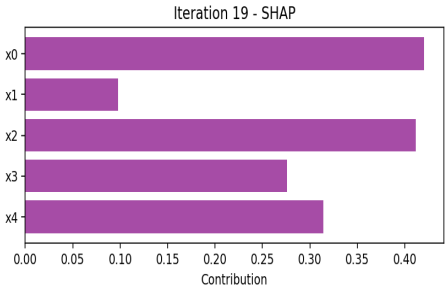
Iteration 18 - SHAP Values

Feature	SHAP Value
x0	0.41635
x1	0.11052
x2	0.42995
x3	0.28184
x4	0.36942

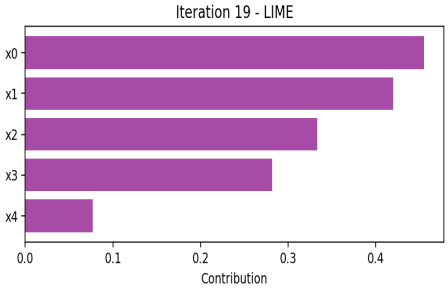
Iteration 18 - LIME Values

Feature	LIME Value
x0	0.46713
x1	0.42817
x2	0.33649
x3	0.26101
x4	0.11622

Iteration 19 - SHAP



Iteration 19 - LIME



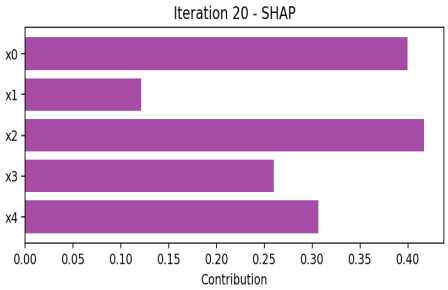
Iteration 19 - SHAP Values

Feature	SHAP Value
x0	0.42014
x1	0.09759
x2	0.41143
x3	0.27570
x4	0.31395

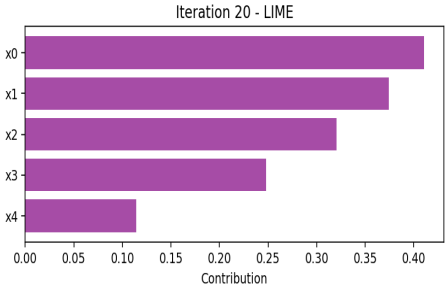
Iteration 19 - LIME Values

Feature	LIME Value
x0	0.45475
x1	0.41970
x2	0.33318
x3	0.28193
x4	0.07724

Iteration 20 - SHAP



Iteration 20 - LIME



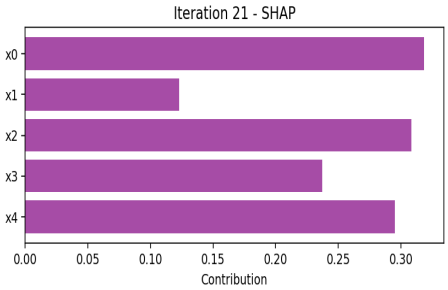
Iteration 20 - SHAP Values

Feature	SHAP Value
x0	0.39982
x1	0.12140
x2	0.41676
x3	0.25985
x4	0.30635

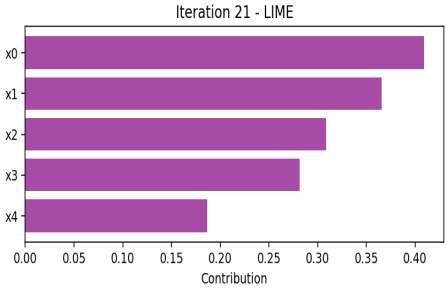
Iteration 20 - LIME Values

Feature	LIME Value
x0	0.41037
x1	0.37420
x2	0.32064
x3	0.24775
x4	0.11414

Iteration 21 - SHAP



Iteration 21 - LIME



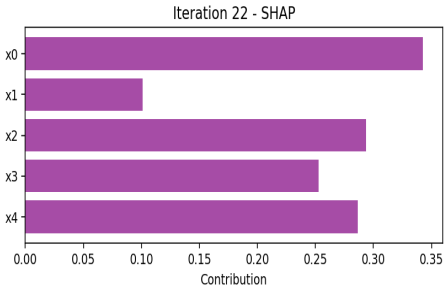
Iteration 21 - SHAP Values

Feature	SHAP Value
x0	0.31840
x1	0.12303
x2	0.30847
x3	0.23698
x4	0.29529

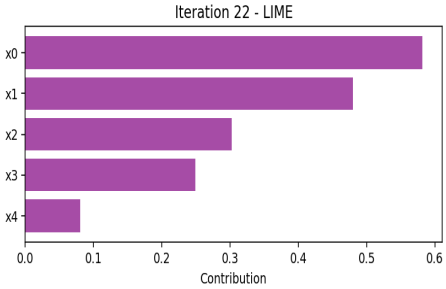
Iteration 21 - LIME Values

Feature	LIME Value
x0	0.40900
x1	0.36541
x2	0.30851
x3	0.28165
x4	0.18640

Iteration 22 - SHAP



Iteration 22 - LIME



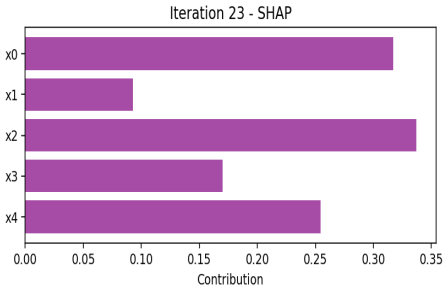
Iteration 22 - SHAP Values

Feature	SHAP Value
x0	0.34249
x1	0.10101
x2	0.29363
x3	0.25277
x4	0.28623

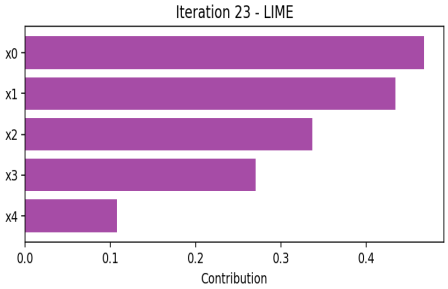
Iteration 22 - LIME Values

Feature	LIME Value
x0	0.58174
x1	0.48001
x2	0.30216
x3	0.24933
x4	0.08014

Iteration 23 - SHAP



Iteration 23 - LIME



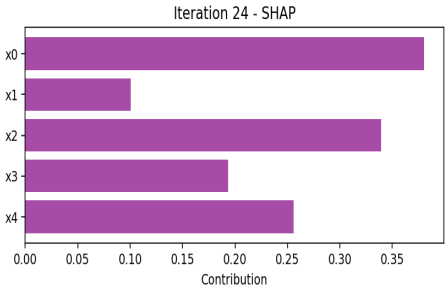
Iteration 23 - SHAP Values

Feature	SHAP Value
x0	0.31699
x1	0.09266
x2	0.33715
x3	0.16998
x4	0.25467

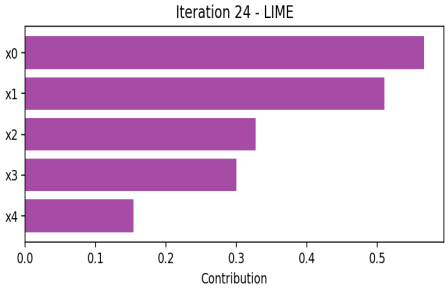
Iteration 23 - LIME Values

Feature	LIME Value
x0	0.46733
x1	0.43411
x2	0.33645
x3	0.27010
x4	0.10776

Iteration 24 - SHAP



Iteration 24 - LIME



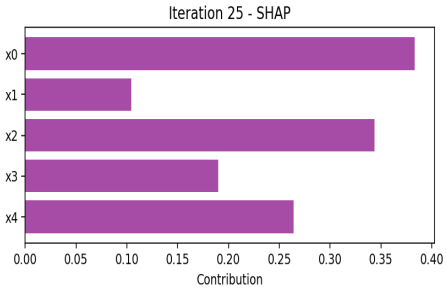
Iteration 24 - SHAP Values

Feature	SHAP Value
x0	0.38003
x1	0.10049
x2	0.33946
x3	0.19340
x4	0.25582

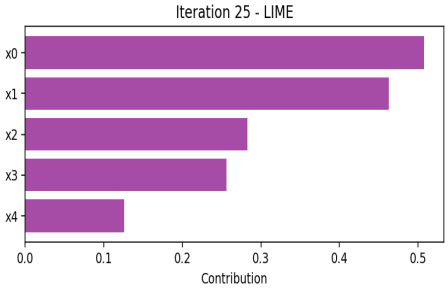
Iteration 24 - LIME Values

Feature	LIME Value
x0	0.56579
x1	0.50939
x2	0.32726
x3	0.29978
x4	0.15385

Iteration 25 - SHAP



Iteration 25 - LIME



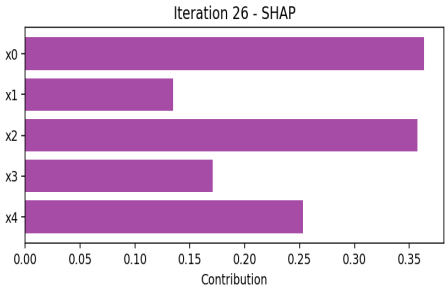
Iteration 25 - SHAP Values

Feature	SHAP Value
x0	0.38356
x1	0.10432
x2	0.34368
x3	0.19010
x4	0.26414

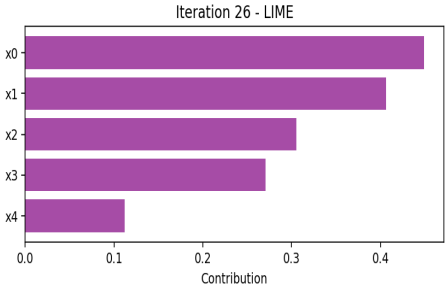
Iteration 25 - LIME Values

Feature	LIME Value
x0	0.50758
x1	0.46287
x2	0.28289
x3	0.25604
x4	0.12589

Iteration 26 - SHAP



Iteration 26 - LIME



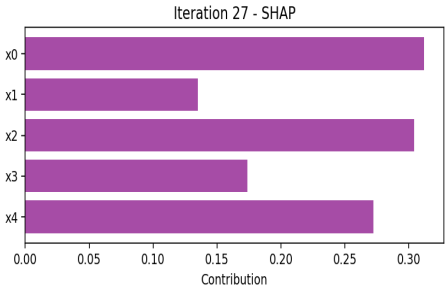
Iteration 26 - SHAP Values

Feature	SHAP Value
x0	0.36330
x1	0.13480
x2	0.35761
x3	0.17096
x4	0.25285

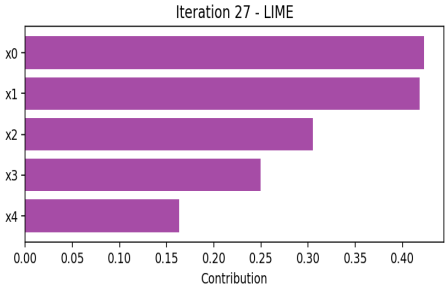
Iteration 26 - LIME Values

Feature	LIME Value
x0	0.44889
x1	0.40644
x2	0.30557
x3	0.27062
x4	0.11160

Iteration 27 - SHAP



Iteration 27 - LIME



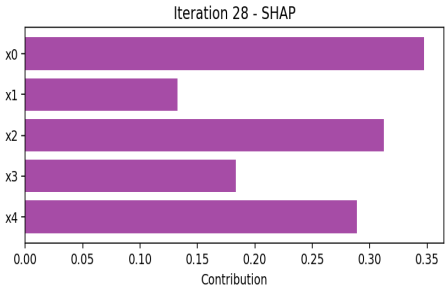
Iteration 27 - SHAP Values

Feature	SHAP Value
x0	0.31180
x1	0.13500
x2	0.30423
x3	0.17359
x4	0.27246

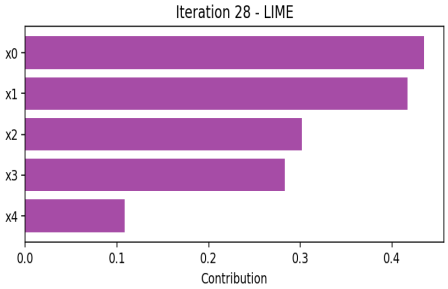
Iteration 27 - LIME Values

Feature	LIME Value
x0	0.42279
x1	0.41823
x2	0.30495
x3	0.24935
x4	0.16293

Iteration 28 - SHAP



Iteration 28 - LIME



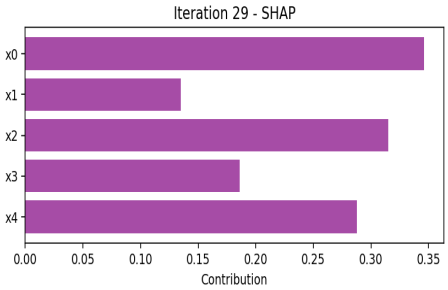
Iteration 28 - SHAP Values

Feature	SHAP Value
x0	0.34716
x1	0.13249
x2	0.31254
x3	0.18324
x4	0.28872

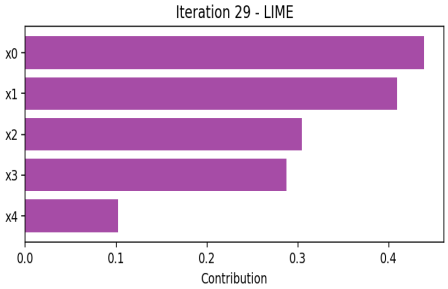
Iteration 28 - LIME Values

Feature	LIME Value
x0	0.43491
x1	0.41726
x2	0.30173
x3	0.28307
x4	0.10833

Iteration 29 - SHAP



Iteration 29 - LIME



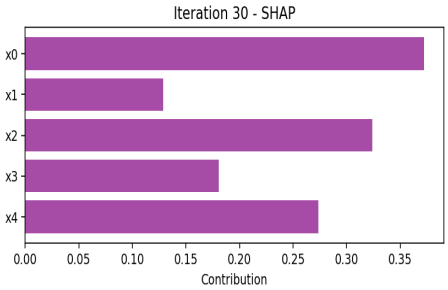
Iteration 29 - SHAP Values

Feature	SHAP Value
x0	0.34592
x1	0.13508
x2	0.31478
x3	0.18594
x4	0.28773

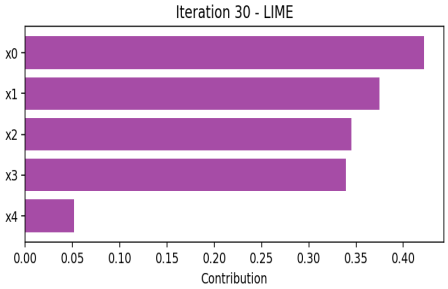
Iteration 29 - LIME Values

Feature	LIME Value
x0	0.43882
x1	0.40907
x2	0.30473
x3	0.28767
x4	0.10192

Iteration 30 - SHAP



Iteration 30 - LIME



Iteration 30 - SHAP Values

Feature	SHAP Value
x0	0.37211
x1	0.12882
x2	0.32387
x3	0.18060
x4	0.27347

Iteration 30 - LIME Values

Feature	LIME Value
x0	0.42168
x1	0.37451
x2	0.34490
x3	0.33908
x4	0.05180